

Daily Sequences Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Binomial & Trinomial Expansion · Pascal's Triangle as a Sequence · The Generalized Binomial Series.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Pascal's triangle and binomial coefficients on sight.

- Write down row 5 of Pascal's triangle (the coefficients of $(x + y)^5$).
- Evaluate $\binom{6}{2}$.
- Write down $\sum_{k=0}^n \binom{n}{k}$ in closed form.
- Expand $(1 + x)^3$ fully.
- Find the coefficient of x^2 in $(1 + x)^6$.
- Row 4 of Pascal's triangle is 1, 4, 6, 4, 1. Is this sequence of five numbers arithmetic? Justify with one check.
- Simplify $\binom{n}{0} + \binom{n}{1}$.
- State the value of $\sum_{k=0}^n (-1)^k \binom{n}{k}$ for $n \geq 1$.
- Find the constant term in $(1 + x)^5$.
- True or false: row n of Pascal's triangle always has exactly $n + 1$ entries.

Section B Manipulation Drills

(≤50 s each)

Substitute into the binomial expansion; extract specific coefficients.

- Prove that $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$ by substituting a value into the expansion of $(1 + x)^n$.

2. Row 10 of Pascal's triangle sums to 2^{10} . If instead you sum only its first four entries, $\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3}$, is the result arithmetic, geometric, or neither, as a rule? Compute the value.
3. Using $x = -1$ in $(1 + x)^n$ ($n \geq 1$), what is $\sum_{k=0}^n (-1)^k \binom{n}{k}$?
- A) 0
B) 1
C) 2^n
D) $(-1)^n$
4. Find the coefficient of x^3 in $(2 + x)^5$.
- A) 10
B) 40
C) 80
D) 20
5. Find the coefficient of x^4 in $(1 - 2x)^6$.
- A) 60
B) -240
C) 240
D) -60
6. Find the constant term in the expansion of $\left(x + \frac{1}{x}\right)^4$.
- A) 4
B) 6
C) 8
D) 1
7. In the expansion of $(1 + x)^n$, the coefficient of x^2 is 21 times the coefficient of x^1 . Find n .
- A) 21
B) 42
C) 43
D) 44
8. The coefficients of x^2 and x^3 in the expansion of $(1 + x)^n$ are equal. Find n .
- A) 4
B) 5

- C) 6
- D) Cannot be determined.
9. Find the coefficient of x^2 in the expansion of $(1 + 2x + 3x^2)^3$.
- A) 15
- B) 18
- C) 21
- D) 24
10. The row-sums of Pascal's triangle are 1, 2, 4, 8, 16, ... Is this sequence of row-sums arithmetic, geometric, both, or neither? State the relevant ratio or difference.

Section C Substitution & Structure(≤ 90 s each)*Structure over computation — what the binomial series is really telling you.*

1. Which of the following statements about partial sums of binomial coefficients is correct?
- A) $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{k-1}$ always equals 2^n , for any $k \leq n$.
- B) No simple closed form exists.
- C) Partial sums of binomial coefficients are always arithmetic in k .
- D) Partial sums of binomial coefficients are always geometric in k .
2. Find the coefficient of x in the expression $(1+x)^0 + (1+x)^1 + \cdots + (1+x)^{40}$. (after TMUA 2019 Paper 1 Q3)
- A) 40
- B) 41
- C) 820
- D) 1640
- E) 1681
3. Find, in terms of n , the coefficient of x^2 in the expression $(1+x)^0 + (1+x)^1 + \cdots + (1+x)^n$. (after TMUA 2019 Paper 1 Q3)
- A) $\binom{n}{3}$
- B) $\binom{n+1}{3}$
- C) $\binom{n+1}{2}$
- D) $2^n - n - 1$
4. In the expansion of $(1+x+x^2)^4$, what is the coefficient of x^3 ?

- A) 12
- B) 16
- C) 20
- D) 24

5. Which of the following is the expansion of $(1 - x)^{-1}$ as an infinite series in x ?

- A) $1 - x + x^2 - x^3 + \dots$
- B) $1 + x + x^2 + x^3 + \dots$
- C) $1 - 2x + 3x^2 - 4x^3 + \dots$
- D) Expansion is not possible.

6. The generalized binomial expansion gives $(1 - x)^{-2} = \sum_{k=0}^{\infty} (k+1)x^k$. Find the coefficient of x^3 .

- A) 3
- B) 4
- C) 6
- D) 10

7. Which of the following is FALSE, for a positive integer n ?

- A) $\sum_{k=0}^n \binom{n}{k} = 2^n$
- B) $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$
- C) $\binom{n}{k} = \binom{n}{n-k}$ for all $0 \leq k \leq n$
- D) $\sum_{k=0}^n k \binom{n}{k} = 2^n$

8. The infinite series $1 + x + x^2 + x^3 + \dots$ equals $(1 - x)^{-1}$. Using $(1 - x)^{-3} = \sum_{k=0}^{\infty} \binom{k+2}{2} x^k$, find the coefficient of x^5 in $(1 - x)^{-3}$.

- A) $\binom{5}{2}$
- B) $\binom{6}{2}$
- C) $\binom{7}{2}$
- D) $\binom{8}{2}$

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant route — find it.

1. Find the coefficient of x^2y^3 in the expansion of $(1 + x + y^2)^5$. (after TMUA 2020 Paper 1 Q13)
 - A) 0
 - B) 10
 - C) 20
 - D) 30
 - E) 60
2. The coefficient of x^2 in $(1 + x + 2x^2)^3$ equals the coefficient of x^2 in $(1 - ax)^3$. Find the possible value(s) of a . (after TMUA Practice Paper 1 Q19)
 - A) $a = \pm\sqrt{3}$
 - B) $a = \pm 3$
 - C) $a = \sqrt{3}$ only
 - D) $a = 3$ only
3. Prove that $\sum_{k=0}^n k \binom{n}{k} = n \cdot 2^{n-1}$ for every positive integer n , by differentiating the binomial expansion of $(1 + x)^n$.
4. Expanding $(1 + x)^n(1 + x)^m$ two different ways — first by combining the factors as $(1 + x)^{n+m}$, then by multiplying the two separate expansions together and comparing the coefficient of x^k on both sides — produces which identity?
 - A) Vandermonde's identity: $\sum_{i=0}^k \binom{n}{i} \binom{m}{k-i} = \binom{n+m}{k}$.
 - B) $\binom{n}{k} + \binom{m}{k} = \binom{n+m}{k}$.
 - C) $\binom{n}{k} \binom{m}{k} = \binom{n+m}{k}$.
 - D) No such identity exists.
5. The coefficients of $(1 - x)^{-1}$ are all 1 (constant in k). The coefficients of $(1 - x)^{-2}$, namely $(k + 1)$, form an arithmetic sequence in k . What kind of sequence, as a function of k , do the coefficients of $(1 - x)^{-3} = \sum_{k=0}^{\infty} \binom{k+2}{2} x^k$ form?
 - A) Arithmetic.
 - B) Geometric.
 - C) Quadratic in k .
 - D) Constant.

“Pascal’s triangle has been doing your multiplication for four hundred years. Let it.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/David Akinyele-Aje/speed-maths](https://github.com/David-Akinyele-Aje/speed-maths)